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Disturbance-Tolerant Quadrotor Control Using a Hybrid LQR and Super-Twisting Sliding Mode Approach

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ABSTRACT

This study proposes a robust cascaded hybrid control architecture combining the Linear Quadratic Regulator (LQR) with Super-Twisting Sliding Mode Control (STSMC) to address the attitude stabilization problem of underactuated quadrotor UAVs. In this framework, the inner LQR loop ensures optimal stabilization, while the outer STSMC loops provide robust reference modification to compensate for uncertainties. To maximize the robustness of the proposed hybrid architecture, the parameters of the STSMC component were globally optimized using the Big Bang-Big Crunch (BB-BC) algorithm. The effectiveness of the proposed strategy is rigorously evaluated on the Quanser 3-DOF Hover system model against two distinct benchmarks: the classical LQR and the optimization-based Model Predictive Control (MPC). Comprehensive simulation scenarios were conducted, including stochastic wind disturbances, time varying thrust coefficients, sudden payload release, and critical motor failure. Additionally, a virtual 3D spatial analysis based on kinematic reconstruction was performed to validate the positional stability of the rotor tips. The comparative results demonstrate that while the MPC controller exhibits a faster transient response compared to the classical LQR, it suffers from significant performance degradation under model mismatches. In contrast, the proposed LQR-STSMC consistently outperforms both benchmarks, achieving error metrics (ISE, IAE, and ITAE) orders of magnitude lower than the competitors. The findings confirm that the hybrid scheme offers superior fault tolerance, faster stabilization, and precise tracking capabilities, making it a highly reliable alternative for safety critical flight operations in uncertain environments.

Keywords: Fault-Tolerant Control, Payload Variation, Parametric Uncertainty, External Disturbances, Big Bang–Big Crunch Optimization, Hybrid Control, Linear Quadratic Regulator, Model Predictive Control, Super-Twisting Sliding Mode Control.

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1. Introduction

Unmanned Aerial Vehicles (UAVs), particularly quadrotors, have attracted significant interest in both civilian and military domains due to their vertical take-off and landing capabilities, high maneuverability, low cost, and structural simplicity [1–2]. Despite these advantages, quadrotors are constrained by limited payload capacity, energy constraints, and high sensitivity to external disturbances [3-4]. Their flight stability and performance are closely tied to the effectiveness of the employed control strategies.

Conventional linear controllers such as Proportional Integral Derivative (PID) and Linear Quadratic Regulator (LQR) are widely used [5], but often fall short under real world conditions involving disturbances, uncertainties, or actuator faults. In this context, Model Predictive Control (MPC) has gained prominence for its ability to handle state constraints and optimize trajectory tracking; however, its heavy reliance on accurate internal models often leads to performance degradation under significant parametric uncertainties [6]. Consequently, more robust strategies such as Sliding Mode Control (SMC), Adaptive Control, and Fault-Tolerant Control (FTC) have been proposed [7-8] to ensure stability against unmodeled dynamics. As quadrotors are inherently underactuated systems, with six degrees of freedom but only four control inputs, they are especially vulnerable to external factors like wind, parametric changes, and sudden mass variations [9-12].

Recent advancements in control theory have introduced sophisticated nonlinear strategies to handle parametric variations and environmental disturbances in aerial systems. For instance, Maaruf et al. [13] proposed a robust architecture combining Adaptive Fractional-Order SMC with Nonsingular Fast Terminal SMC and a finite time disturbance observer. Similarly, Fikadu Assefa et al. [14] introduced an Adaptive Neuro-Fuzzy Inference System based SMC to effectively manage external disturbances and parameter variations in quadcopter UAVs. Addressing the specific challenges of formation flight, Nguyen and Pitakwachara [15] developed a Non Singular Terminal Super-Twisting SMC, highlighting that integrating super-twisting algorithms with disturbance observers significantly enhances response speed.

Beyond sliding mode variations, hybrid and optimization-based approaches have also been explored. Mahmoud and Maaruf [16] introduced a multilevel control strategy utilizing Radial Basis Function Neural Networks. Addressing the complexity of parameter tuning, Maaruf et al. [17] applied metaheuristic algorithms specifically Improved Grey Wolf and Whale Optimization to optimize backstepping controllers. Similarly, evolutionary techniques such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) have been successfully employed to optimize Sliding Mode Controller gains for unstable nonlinear systems, including magnetic levitation trains and suspension systems [18-19].

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Furthermore, robust control applications extend to the broader field of robotics and mechatronics. Guo et al. [20] demonstrated the efficacy of global fast terminal SMC for manipulators. Recent literature also highlights the implementation of Optimized Backstepping Fuzzy SMC for mobile manipulators [21] and Adaptive Fuzzy Super-Twisting Sliding Mode Control (STSMC) for robotic arms [22-23]. In the biomedical domain, advanced strategies like Neural Network based STSMC and Fuzzy SMC have been utilized for lower limb prostheses [24] and dynamic walking assistance in exoskeletons [25].

In parallel with these developments, recent studies have investigated other hybrid configurations to further enhance tracking accuracy. For example, [26] proposed an improved trajectory tracking control using a Novel SMC with a Fuzzy PID Surface, while [27] utilized a Backstepping Fuzzy SMC for mobile robots. Additionally, [28] developed an Improved Nonsingular Adaptive STSMC to address system uncertainties.

While the reviewed advanced strategies spanning from adaptive and fractional order sliding modes to neural network-based and optimization driven architectures demonstrate impressive tracking capabilities, they often introduce significant implementation challenges. These methodologies typically entail high computational complexity due to the necessity of executing iterative optimization algorithms online, training neural structures, or designing intricate adaptation laws to handle fractional dynamics. Seeking a balance between computational efficiency and robust performance, this study proposes a hybrid control approach, combining LQR with Super-Twisting Sliding Mode Control (LQR-STSMC). The selection of the LQR-STSMC hybrid architecture is driven by the need to balance energy optimality with robust disturbance rejection, a trade off that single method controllers often fail to achieve. Linear controllers like LQR are renowned for their ability to minimize control effort and steady state error around the operating point, yet they degrade significantly under nonlinear disturbances. Conversely, nonlinear robust strategies, particularly conventional SMC, offer excellent insensitivity to uncertainties but typically suffer from the “chattering” phenomenon, which causes high frequency mechanical wear and energy waste. To overcome these limitations, this study proposes a synergistic approach. The LQR component is employed to dictate the nominal system dynamics, ensuring smooth and energy-efficient tracking. Simultaneously, the STSMC a second order sliding mode algorithm is integrated as a corrective term. Unlike standard SMC, the STSMC provides finite time convergence and significantly attenuates chattering by hiding the discontinuous sign function under an integral. This specific combination allows the quadrotor to retain the optimality of linear control in calm conditions while instantly leveraging the high robustness capabilities of STSMC when subjected to external perturbations or failures.

Distinct from the fuzzy or fully adaptive architectures referenced above, the philosophy of the proposed LQR-STSMC framework lies in its structural simplicity. Instead of relying on fuzzy approximations, the method

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integrates LQR as the primary baseline controller to ensure energy optimality in nominal conditions, while STSMC serves strictly as an auxiliary corrective layer. The proposed method is evaluated through four simulation scenarios: external wind disturbance, time varying thrust coefficients, motor failure, and sudden payload release. A comparative analysis against the classical LQR approach demonstrates the hybrid controller's enhanced stability and tracking performance under nominal and disturbed conditions.
The main contributions of this study to the existing literature can be summarized as follows:

Novel Optimized Hybrid Architecture: A cascaded control structure combining the optimal stabilization of LQR with the robustness of STSMC is proposed. Unlike conventional heuristic tuning methods, the parameters of the STSMC component are globally optimized using the BB-BC algorithm to maximize disturbance rejection capability.
Comprehensive Benchmarking against MPC: Distinct from many studies that limit comparisons to PID or LQR, this research rigorously evaluates the proposed method against the optimization-based MPC, providing a critical analysis of the trade-offs between linear, model-based, and sliding mode control strategies.
Performance Under Critical Failure Modes: The robustness of the controller is tested not only under standard disturbances (wind) but also under severe safety critical scenarios, including sudden payload release and partial motor failure, demonstrating superior fault tolerance where conventional methods fail.
3D Kinematic Validation: A virtual 3D spatial analysis of the rotor tip trajectory is introduced to visually and quantitatively verify the positional stability and oscillation envelope of the system, offering a deeper insight into the physical implications of the control signals.

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2. Review of Operational Scenarios and Control Challenges

To establish a comprehensive validation framework for the proposed hybrid controller, it is essential to analyze the specific operational challenges prevalent in UAV flight missions. While the Introduction provided a broad overview of control strategies, this section focuses on the state of the art regarding the four critical scenarios selected for this study: environmental disturbances, parametric uncertainties, actuator faults, and payload variations.

Scenario 1: Wind Disturbance Mitigation

Atmospheric disturbances, particularly wind gusts, represent a primary source of instability for lightweight quadrotors, often inducing significant tracking errors [29-30]. Given the stochastic and nonlinear nature of wind dynamics, accurate aerodynamic modelling and robust control synthesis are critical [31]. For instance, Jeon et al. [32] integrated aerodynamic effects into the quadrotor dynamic model using blade element theory, validating their approach experimentally. Similarly, Svacha et al. [33] and Perozzi et al. [34] incorporated aerodynamic force estimations directly into the control loop, utilizing gain scheduled SMC to enhance tracking accuracy and reduce chattering. Furthermore, Raza et al. [35]

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demonstrated that a hybrid PD-adaptive backstepping structure could significantly minimize position errors under wind loading. These studies collectively emphasize the necessity for aerodynamic aware and disturbance resilient control architectures.

Scenario 2: Handling Time Varying Thrust Coefficients

Time varying parametric uncertainties, stemming from factors such as motor heating or battery voltage depletion, pose severe challenges to flight stability [36]. To mitigate these effects, Mu and Zhang [37] proposed a neural network based robust controller designed specifically for coupled, time varying systems. Addressing delay induced instabilities, Liu et al. [38] introduced a hierarchical control structure capable of managing uncertainties effectively. Additionally, Hassani et al. [39] developed a hybrid SMC approach featuring adaptive gain adjustment, which achieved high robustness and fast convergence despite parameter shifts. Consequently, the literature underscores the importance of adaptive and delay tolerant techniques for operating in uncertain parameter spaces.

Scenario 3: Tolerance to Motor Failure

Actuator failure constitutes a critical safety risk in quadrotors, as it induces severe asymmetry in thrust and moments. FTC strategies aim to sustain flight capability using the remaining healthy actuators [40]. Research by Lee et al. [41] demonstrated the feasibility of maintaining control during a single motor failure by dynamically shifting the center of gravity. Betancourt et al. [42] introduced an observer-based architecture designed to accommodate both critical and non-critical actuator faults. Furthermore, Nan et al. [43] utilized Nonlinear MPC for stable maneuvering under thrust saturation limits, while Ke et al. [44] proposed a seamless compensator-based controller requiring no logical switching. These approaches highlight the necessity of intrinsic fault tolerance in modern UAV control systems.

Scenario 4: Adaptation to Payload Variation

Sudden variations in payload mass drastically alter the system’s inertial properties and center of gravity, demanding rapid adaptation from the controller [45]. Sankaranarayanan et al. [46] addressed this by proposing a robust controller specifically for load carrying missions subject to varying constraints. Similarly, Dhiman et al. [47] introduced a hybrid PID-PD controller optimized for cooperative transport tasks, while Kumar and Dewan [48] designed a PID-SMC scheme to ensure robustness against mass variations. The recurring theme in these studies is the critical need for dynamic adaptation mechanisms to handle abrupt changes in system dynamics.

In this context, the proposed LQR-STSMC controller combines the optimality of LQR with the robustness of sliding mode techniques. Unlike many theoretical hybrid approaches, this method is validated through diverse simulation scenarios including wind, parametric uncertainty, actuator failure, and mass variation demonstrating superior transient and steady state performance and offering both practical relevance and a meaningful contribution to the literature.

Conclusion of the Review While the cited literature offers effective solutions for individual challenges ranging from wind mitigation to fault tolerance few studies present a unified framework capable of addressing

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all these scenarios simultaneously with a computationally efficient structure. In this context, the proposed LQR-STSMC controller is designed to bridge this gap. By combining the energy optimality of LQR with the robust disturbance rejection of STSMC, this study aims to validate a versatile control strategy that maintains superior transient and steady-state performance across this diverse set of operational conditions.

3. Materials and Methods

3.1. System Description and Scope

The 3-degree-of-freedom (3-DOF) hover training facility comprises a structure equipped with four propellers. This system is mounted on a pivot joint that allows rotation around the yaw, roll, and pitch axes [49]. The base of the plant is fixed to a workbench and incorporates slip rings to enable free movement along the yaw axis with minimal friction.

Each propeller generates thrust, which is used to control the roll and pitch angles of the system. As the propellers rotate, the resulting torque induces motion around the yaw axis. In a controlled environment, when the forces generated by the propellers are balanced, the net torque becomes zero, thereby stabilizing the system.

For this study, the model provided in the manufacturer's manual was used [49]. When a positive voltage is applied to any of the motors, buoyancy is generated, which causes the propulsion system to rise. The front and rear motors, controlled by voltages U_f and U_b, govern the motion along the tilt and yaw axes. Meanwhile, the side motors, controlled by U_r and U_l, are responsible for motion along the roll and yaw axes [49].

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3.1.1. State Space Model

Following the linearization of the system's mathematical model, a state-space representation of the 3-DOF four rotor system was developed. This model incorporates four inputs and three outputs, accurately capturing the dynamics of the system. The detailed mathematical formulation of the model is presented in Equations (1-9) [49].

$$\dot{X} = AX + \quad (1)$$

$$Y = CX + \quad (2)$$

State vector;

$$X^T = [Y \;\; P \;\; R \;\; \dot{Y} \;\; \dot{P} \;\; \dot{R}] \quad (3)$$

Control vector;

$$U^T = [U_f \;\; U_b \;\; U_r \;\; U_l \quad (4)$$

Output vector;

$$Y^T = \quad (5)$$

Here;

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U_f: Front motor voltage, U_b: Back motor voltage, U_r: Right motor voltage, U_l : Left motor voltage.

Matrices A, B, C and D of the state space model;

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{-K_t}{J_y} & \frac{-K_t}{J_y} & \frac{K_t}{J_y} & \frac{K_t}{J_y} \\ \frac{K_f.L}{J_p} & -\frac{K_f.L}{J_p} & 0 & 0 \\ 0 & 0 & \frac{K_f.L}{J_r} & -\frac{K_f.L}{J_r} \end{pmatrix} (7)$$

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} (8)$$

$$D = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} (9)$$

The parameters of the Quanser 3-DOF quadrotor were utilized to simulate the proposed control method [49]. Detailed in Table 1, these parameters include key physical and control related properties such as motor constants, propeller dimensions, system inertia, and friction coefficients, which are critical for accurately modelling and controlling the system's behavior.

Understanding these parameters is essential for designing effective control strategies to manage the dynamics of the system along the yaw, roll, and pitch axes. In this study, Table 1 serves as a reference for configuring the system and ensuring that the mathematical model aligns closely with the real world behavior of the plant [49].

Table 1. Parameters of the 3-DOF hover system

Symbol	Value	Unit
K_t	0.0036	N.m/V
K_f	0.1188	N/V
L	0.1969	m
J_y	0.1104	$Kg.m^2$
J_p	0.0552	$Kg.m^2$
J_r	0.0552	$Kg.m^2$

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3.2. Linear Quadratic Regulator (LQR)

The LQR is a widely used control approach for linear dynamic systems. This method aims to develop an optimal control strategy for systems defined by linear differential equations. The primary objective of the LQR method is to design a control law that ensures the desired dynamic performance of the system while minimizing a predefined cost function.

This cost function is typically formulated to balance the deviation of system states from their desired values and the control effort applied, with the goal of minimizing the total cost over time.

The LQR control law is generally defined as shown in Equation (10) [50].

$$U = - \quad (10)$$

Where, K represents the optimal state feedback gain matrix. The objective of the LQR method is to minimize the quadratic cost function defined in Equation (11) [50]:

$$J = \int_0^\infty (x^T.Q.x + u^T.R.u)dt \quad (11)$$

In this equation, Q represents the weighting matrix for the state vector, while R denotes the weighting matrix for the control input. These weights are designed to establish the desired trade-off between system performance and control effort.

However, while LQR offers optimal performance for linear systems under nominal conditions, it does not always guarantee sufficient performance under real world conditions such as external disturbances and parametric uncertainties. Therefore, although the high precision of LQR is advantageous, it has become increasingly common to employ LQR in a hybrid structure alongside sliding surface-based control methods that offer greater robustness.

In this study, the stable control capabilities of LQR are combined with the robustness of the STSMC technique to construct a more resilient framework capable of handling external disturbances and uncertainties effectively.

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3.3. Model Predictive Control (MPC)

In this study, the MPC method, one of the model-based control approaches, is employed for feedback control design. MPC is a control algorithm that optimizes control inputs in real time by utilizing a mathematical model capable of predicting the future behavior of the system, with the objective of minimizing a predefined cost function. Its ability to incorporate system constraints during decision-making and its suitability for multi-input multi-output (MIMO) systems are among the key features that make MPC highly advantageous in industrial applications [51].

The system used in this study is represented by a linearized MIMO state-space model with four inputs and three outputs. The controller was implemented in real time within the Simulink environment, and the entire control process was carried out using the MATLAB/Simulink interface. The MPC architecture was configured as follows:

- Sampling time (Ts): 0.002 seconds

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- Prediction horizon (p): 100 steps
- Control horizon (m): 15 steps
- Manipulated variable (MV) limits: between 0 and 24 V

During the control process, desired reference signals were applied to the system outputs, and the MPC algorithm was used to drive the system responses to follow these references. The state-space model was used throughout the control design process, and the physical constraints of the system were considered by imposing limits on the control inputs.

The MPC algorithm operates based on the moving horizon principle. At each sampling instant, the current state of the system is measured, and the future behavior of the system is predicted over the defined prediction and control horizons. Only the first control input in the optimized sequence is applied, and the process is repeated at the next sampling instant to adapt to the system dynamics in real time. Through this structure, MPC ensures optimal reference tracking and effectively handles physical system constraints by generating smooth control signals based on the internal predictive model [51].

3.4. Super-Twisting Sliding Mode Control (STSMC)

STSMC is a second order sliding mode control algorithm specifically designed for systems with a relative degree of one. This method generates a continuous control signal that drives both the sliding variable and its derivative to zero in finite time. STSMC demonstrates superior performance compared to conventional sliding mode control techniques, particularly in reducing undesirable chattering effects caused by discontinuous control inputs [52].

A key feature of this algorithm is the decomposition of the control signal into two distinct components: the first is derived from the time derivative of a discontinuous function dependent on the sliding surface, while the second is a continuous complementary term with respect to the surface. This structure enhances the system's robustness against disturbances and model uncertainties, while also improving stability and overall tracking accuracy. Moreover, the spiral convergence of the system trajectories toward the sliding surface contributes to improved transient response behavior.

The version of STSMC employed in this study is also referred to in the literature as Modified STSMC. The corresponding control laws and the mathematical formulation of the algorithm are presented in Equations (12)–(13) [53].

$$e(t) = Q_{ref}(t) - \quad (12)$$

$$\dot{e}(t) = \dot{Q}_{ref}(t) - \dot{Q}(t) \quad (13)$$

$Q(t)=[Y(t)\ P(t)\ R(t)]^T$ where, $Q(t)$ represents the actual angle vector consisting of the Yaw, Pitch, and Roll angles, and Q_{ref} denotes the reference angle vector. Since the control design strategy is identical for each axis, the formulation is presented for a generic axis i, where $i \in \{Yaw, pitch, roll\}$. First, the sliding surface $s_i(t)$ for the i-th axis is defined as a weighted sum of the tracking error $e_i(t)$ and its derivative: [53]

$$s_i(t) = \lambda_i . e_i(t) + \dot{e}_i(t) \quad (14)$$

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Where, Equations (14) Where $\lambda_i>0$ determines the slope of the sliding surface.
The control input $u(t)\in\mathbb{R}^3$ is defined based on a smooth STSMC strategy as follows Equation (6) [53]:

$$u(t) = - k_{1,i}.|s_i|^{\frac{1}{2}}.\tanh\left(\frac{s_i}{\varnothing}\right) - \int_0^t k_{2,i}.\tanh\left(\left(\frac{s_i(\tau)}{\varnothing}\right)\right)d\tau \quad (15)$$

Where, $k_{1,i}$ and $k_{2,i}$ are the positive adaptive gains, and $\varnothing$ (set to 0.99 in this study) is the scaling parameter that defines the boundary layer width for the smooth transition.
Given that the system requires continuous control signals, the hyperbolic tangent function tanh is employed as a smooth approximation of the conventional discontinuous sign function to mitigate chattering. The control law presented in Equation (15) comprises two primary components: the first term ensures rapid convergence of the system toward the sliding surface, while the second guarantees stability and maintains the system on the manifold. This structure provides superior transient performance and robust resilience against model uncertainties and external disturbances. Furthermore, utilizing the hyperbolic tangent function in the sliding surface definition yields smoother and more stable control signals compared to traditional SMC approaches.

3.5. Integrated Use of LQR and STSMC: Robust Virtual Reference Generation

In this study, a cascaded hybrid control framework is proposed to enhance the robustness of the Quanser 3-DOF hover system. Unlike conventional additive control architectures where auxiliary signals are directly added to the control input, this strategy utilizes the STSMC as an outer loop robust compensator. The fundamental scientific principle here is the separation of nominal performance and robustness: the LQR is dedicated to optimal stabilization based on the known linear dynamics, while the STSMC is tasked with estimating and rejecting “lumped disturbances” (including external wind, parametric uncertainties, and unmodeled dynamics).

The control architecture operates in a hierarchical dual loop structure:

Outer Loop (Robust Supervisor): The STSMC algorithm monitors the system’s output states. Instead of generating a direct torque, it functions as a “virtual controller” that calculates the necessary state deviation required to counteract the acting disturbances.
Dynamic Reference Modification: This corrective signal acts as a virtual offset. It is subtracted from the user-defined nominal reference to generate a robust modified reference (r_{mod}) Essentially, the STSMC shifts the equilibrium point perceived by the linear controller to balance out the external forces.
Inner Loop (Optimal Tracking): The LQR controller acts as a high-performance tracker for this new dynamic reference (r_{mod}), By minimizing the error between the system states and this modified trajectory, the LQR

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implicitly applies the precise control effort needed to maintain the actual system at the desired setpoint, effectively nullifying the disturbance.

First, the sliding surface $s_i(t)$ for i-th axis is defined as a weighted sum of the tracking error $e_i(t)$ and its derivative: [53]

$$s_i(t) = \lambda_i . e_i(t) + \dot{e}_i(t) \quad (16)$$

Where $\lambda_i > 0$ determines the slope of the sliding surface. To ensure chattering free operation suitable for the UAV actuators, the corrective signal $u_{comp,i}(t)$, is generated using the continuous Super-Twisting algorithm in the integral form: [53].

$$u_{comp,i}(t) = - k_{1,i} . |s_i|^{\frac{1}{2}} . \tanh\left(\frac{s_i}{\varnothing}\right) - \int_0^t k_{2,i} . \tanh\left(\left(\frac{s_i(\tau)}{\varnothing}\right)\right) d\tau \quad (17)$$

Where, $k_{1,i}$ and $k_{2,i}$ are the positive adaptive gains, and $\varnothing$ (set to 0.99 in this study) is the scaling parameter that defines the boundary layer width for the smooth transition.

This compensation signal is then used to shift the nominal reference trajectory ($r_{nom,i}(t)$), yielding the robust modified reference $r_{mod,i}(t)$:

$$r_{mod,i}(t) = r_{nom,i}(t) - u_{comp,i}(t) \quad (18)$$

Finally, the LQR controller generates the control input $u_i(t)$ by driving the system states $x_i(t)$ to track this modified reference. The control law is defined as:

$$u_i(t) = K_{LQR} \times (r_{mod,i}(t) - x_i(t) \quad (19)$$

In this configuration, the LQR gain matrix K_{LQR} processes the deviation from the robust virtual reference. Consequently, while the LQR maintains its optimal feedback characteristics, the STSMC effectively “tricks” the linear controller into applying the necessary offset forces to reject external disturbances and steady-state errors.

The proper selection of parameters (λ_i, $k_{1,i}$, $k_{2,i}$) as a direct impact on the fine tuning of the sliding surface dynamics and the transient response performance of the system. Therefore, it is recommended that these parameters be optimized according to the specific application.

The efficacy of the sliding mode dynamics and the transient response of the system are intrinsically linked to the precise selection of the controller parameters (λ_i, $k_{1,i}$, $k_{2,i}$). Given that manual tuning is often suboptimal, identifying application specific values is essential. To address this, the present study utilizes the Big Bang-Big Crunch (BB-BC) meta-heuristic algorithm to minimize the ITAE, thereby optimizing the control gains. BB-BC algorithm was selected due to its low computational complexity and high convergence speed compared to other evolutionary algorithms such as GA or PSO [54]. Its unique ‘center of mass’ contraction mechanism allows for rapidly narrowing the search space to find the global optimum with minimal computational overhead, making it highly efficient for tuning control parameters. The proposed method shares conceptual similarities with GA, particularly in its initialization phase. It commences with the generation of a stochastic initial population; a process termed the “Big

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Bang" phase. Following this expansion, the algorithm enters the "Big Crunch" phase, which functions as a convergence operator. This operator aggregates the dispersed population into a single representative output, defined as the center of mass (x^c). In this calculation, the "mass" is assigned as the inverse of the fitness function value to prioritize optimal solutions. The center of mass is computed according to the formulation established [55]:

The execution of the BB-BC algorithm involves a series of sequential steps outlined below:

Step 1: An initial population of size N is created by randomly positioning candidate solutions within the feasible limits of the search domain.

Step 2: Following initialization, the performance of every candidate is assessed by calculating its fitness value based on the chosen objective function.

Step 3:

$$x^c = \frac{\sum_{i=1}^{N} \frac{1}{f^i} x^i}{\sum_{i=1}^{N} \frac{1}{f^i}} \quad (20)$$

Here

x^c represents the center of mass, x^i the candidate point, N the number of populations, f^i the objective function value of the candidate point.

Find the center of mass according to Equation (20) [55].

Step 4: The new generation of candidates is calculated by perturbing the center of mass with a normal random number. The magnitude of this random perturbation is inversely proportional to the iteration count, ensuring that the search radius shrinks over time. This generation process is defined by Equation (21) [55].

$$x^{\text{new member}} = x^c + \frac{l.r}{k} \quad (21)$$

Here,

$x^{\text{new member}}$: Member of the new generation,

x^c: Stands for center of mass,

L: Defined upper value,

R: Random number value,

K: Number of iterations

Step 5:

The process repeats from Step 2 until a stopping criterion such as a maximum number of iterations or a convergence threshold is satisfied.

The optimization framework aims to minimize performance metrics such as ITAE, overshoot, settling time, and control effort. By utilizing the best candidate solutions to narrow the search space, the algorithm accelerates the convergence process and terminates upon reaching a predefined error tolerance. The final parameters (λ_i, $k_{1,i}$, $k_{2,i}$) are validated to ensure stability and practical applicability [54].

The specific configuration of the BB-BC algorithm employed in this study is detailed as follows:

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- Algorithm Configuration: The population size was set to N=20 with a maximum iteration count of 100 to ensure a balanced convergence speed.
- Robustness Check: Each optimization trial was executed three independent times to verify the consistency of the obtained fitness values.
- Design Variables: The optimization process focused on simultaneously tuning the STSMC parameters (λ_i, $k_{1,i}$, $k_{2,i}$) for the three rotational axes.
- Cost Function: The algorithm was driven to minimize the ITAE criterion as the primary cost function.
- Search Space: The search domain for (λ_i, $k_{1,i}$, $k_{2,i}$) was strictly defined, with lower and upper bounds selected to ensure system stability during the search process.
- Implementation: The entire procedure was automated using the MATLAB/Simulink interface, allowing for rapid parameter determination.

Consequently, the proposed hybrid architecture establishes a symbiotic relationship between linear optimality and nonlinear robustness. By leveraging the LQR for nominal stabilization and the STSMC for dynamic reference modification as visually depicted in the block diagram of the "Robust Control via Dynamic Reference Modification" strategy shown in Figure 1 the control framework effectively decouples the disturbance rejection task from the trajectory tracking objective. This integration not only mitigates the inherent chattering phenomenon common in conventional sliding mode control but also ensures high-precision tracking for the underactuated and strongly coupled dynamics of the Quanser 3-DOF system, proving its efficacy against both parametric uncertainties and exogenous perturbations.

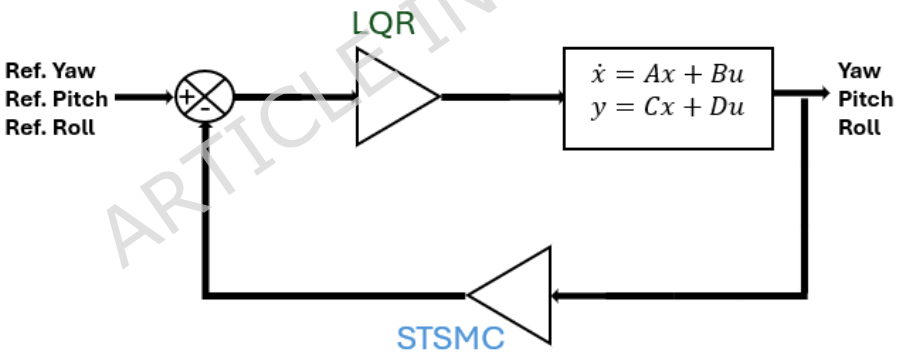

Figure 1. Robust Control via Dynamic Reference Modification

3.5.1. Stability Analysis

The stability of the proposed cascaded control architecture is analyzed by examining the interaction between the asymptotic stability of the inner LQR loop and the practical stability of the outer STSMC loop.

a) Stability of the Inner Loop (LQR)

The inner loop dynamics are governed by the linear system $\dot{x} = Ax(t) + Bu(t)$. The LQR controller generates the control input vector u(t) to track the modified reference vector $r_{mod}(t)$ provided by the STSMC.

The closed-loop dynamics are given by:

$$\dot{x}(t) = (A - BK_{LQR})x(t) + BK_{LQR}r_{mod}(t) \quad (22)$$

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Since the gain matrix K_{LQR} is derived to minimize a quadratic cost function, the closed-loop system matrix $A_{cl} = (A - BK_{LQR})$ is guaranteed to be Hurwitz. This ensures that the real parts of all eigenvalues are negative. Consequently, for any bounded reference signal $r_{mod}(t)$ generated by the outer loop, the system states x(t) remain Bounded-Input Bounded-Output (BIBO) stable across all axes.

b) Stability of the Outer Loop (STSMC)

The outer loop consists of three decoupled STSMC structures. Since the control design is identical for each rotational axis, the stability analysis is presented for a generic axis i. where $i \in \{yaw, pitch, roll\}$.

The sliding variable dynamics for the i-th axis are given by:

$$s_i = -k_{1,i}.|s_i|^{\frac{1}{2}}.\tanh\left(\frac{s_i}{\varnothing}\right) + v_i + d_i \quad (23)$$

$$v_i = -k_{2,i}\tanh\left(\frac{s_i}{\varnothing}\right) \quad (24)$$

To analyze the stability, we consider a Lyapunov function candidate $V_i(z_i) = z_i^T P_i z_i$ for the i-th axis, based on the standard Super-Twisting Algorithm proof. However, the use of the continuous $\tanh(s_i/\varnothing)$ function instead of the discontinuous $\mathrm{sgn}(s_i)$ introduces a perturbation term. The time derivative of the Lyapunov function satisfies:

$$\dot{V}_i(z_i) \leq -\tau_i V_i^{\frac{1}{2}}(z_i) + \delta_i(\varnothing) \quad (25)$$

Where, $\delta_i(\varnothing) > 0$ is a residual term arising from the smooth approximation. This inequality implies that the trajectories for axis i do not converge to zero but are guaranteed to enter and remain within a bounded set:

$$V_i \leq \left(\frac{\delta_i}{\tau_i}\right)^2 \quad (26)$$

Global stability conclusion since the stability condition holds for each axis $i \in \{Yaw, pitch, roll\}$ individually, the total system state in the outer loop is Practically Stable (Globally Uniformly Ultimately Bounded). Consequently, the modification signals $u_{comp,i}$ remain bounded, which ensures that the modified reference vector r_{mod} fed to the inner LQR loop is bounded. Combining this with the BIBO stability of the LQR loop, the entire cascaded closed-loop system is stable.

4. Simulation Studies

In order to evaluate the tracking performance of the system under more realistic and dynamic conditions, reference signals with specific amplitudes and frequencies were applied to each rotational axis. Specifically, a reference signal with an amplitude of ±5° and a frequency of 0.04 Hz was applied for the yaw angle, ±4° and 0.1 Hz for the pitch angle, and ±4° and 0.08 Hz for the roll angle. These specific trajectory parameters were selected to align with standard benchmarks commonly utilized in the literature [56], ensuring a fair and consistent comparison with existing studies. The system's tracking performance was then analyzed based on its response to these inputs.

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Scenario 1: Evaluation of Control Performance Under Wind Disturbance

In the scenario, the robustness of the proposed architecture is rigorously evaluated through a comparative analysis involving three distinct control strategies: the classical LQR, the optimization-based MPC, and the proposed hybrid framework. While MPC serves as a high-performance benchmark representing modern predictive control standards, LQR constitutes the linear baseline. The primary objective of this section is to assess the system's resilience against stochastic external perturbations (simulating wind gusts) and to demonstrate the superior disturbance rejection and transient recovery capabilities of the proposed LQR-STSMC method compared to both linear and optimization-based approaches.

In the simulation environment, external disturbance forces were applied along all three axes to emulate a constant wind effect: F_x=0.3N, F_y=0.2N, F_z=0.1N. These disturbances influence the system's position and attitude dynamics, and the controller's ability to maintain stability and preserve the quality of the transient response under such conditions was investigated.

The simulations were supported by time-domain plots of the system's position and angular responses, and the behavior of the control signals was also analyzed to provide insight into control effort and system robustness.

Table 2. Performance Metrics (ISE, IAE, ITAE) of LQR, MPC and LQR-STSMC Controllers Under Wind Disturbance

Axis	Control	ISE	IAE	ITAE
Yaw	LQR	0.0134	0.3023	3.0758
	MPC	0.0091	0.2237	2.4993
	LQR-STSMC	**0.0015**	**0.1716**	**1.7930**
Pitch	LQR	0.0087	0.1499	1.3634
	MPC	0.0029	0.1795	1.8913
	LQR-STSMC	**0.0002**	**0.0407**	**0.4238**
Roll	LQR	0.0101	0.1715	1.8150
	MPC	0.0032	0.1776	1.9223
	LQR-STSMC	**5.383e-5**	**0.0088**	**0.0834**

The quantitative performance assessment, detailed in Table 2, presents a comprehensive comparison of the LQR, MPC, and proposed LQR-STSMC controllers based on ISE, IAE, and ITAE metrics. A holistic analysis across the Yaw, Pitch, and Roll axes reveals a consistent performance hierarchy, where the proposed hybrid architecture demonstrates distinct superiority over both the linear (LQR) and optimization-based (MPC) benchmarks.

Regarding the Integral Squared Error (ISE), which penalizes large transient deviations, the LQR-STSMC achieved the lowest values across all axes. This indicates that the proposed method is significantly more

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effective at suppressing high-amplitude oscillations and peak errors caused by wind gusts compared to LQR and MPC.

Similarly, the improvements in Integral Absolute Error (IAE) and Integral Time Absolute Error (ITAE) provide insight into the steady-state performance. While the MPC controller generally exhibits competitive tracking capabilities relative to the classical LQR, the LQR-STSMC outperforms both methods by a substantial margin in these metrics. The drastic reduction in ITAE, in particular, confirms that the hybrid controller ensures rapid convergence and effectively eliminates long term steady-state errors.

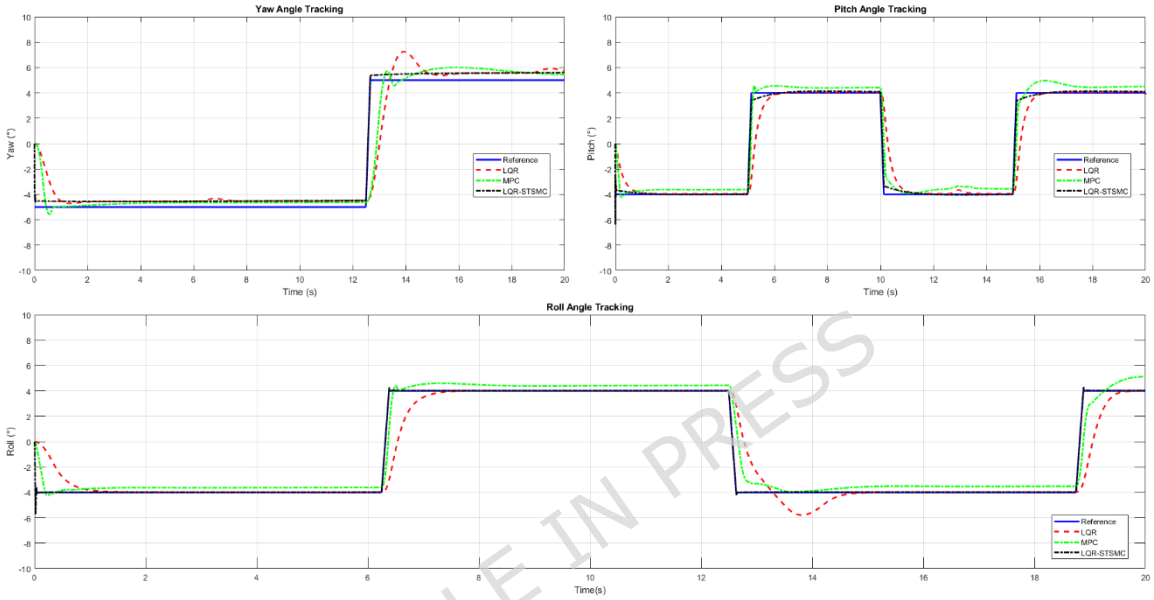

Figure 2. Time Responses of Attitude Angles Under Wind Disturbance

As depicted in Figure 2, although the MPC controller provides a rapid initial response, it exhibits noticeable steady-state offsets under wind disturbance. Conversely, the proposed LQR-STSMC architecture effectively eliminates these errors, delivering the most robust performance with significantly lower oscillation amplitudes and shorter settling times compared to both LQR and MPC benchmarks.

Scenario 2: Time Varying Thrust Coefficient (Parametric Uncertainty)

In this scenario, a case involving time varying thrust (K_f) and torque (K_t) coefficients is modeled to represent parametric uncertainty in the system. This variation is associated with a time dependent decrease in motor voltage, specifically from 24 V to 20 V, simulating realistic conditions such as battery depletion during prolonged missions. The aim is to examine how such a condition affects the system dynamics.

It is assumed that both the thrust and torque coefficients decrease proportionally to the square of the battery voltage. This scenario evaluates the extent to which the proposed controllers can maintain stability and tracking performance under time dependent parameter uncertainties.

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Table 3. Performance Comparison of LQR, MPC and LQR-STSMC Controllers Under Time-Varying Thrust Coefficients (ISE, IAE, ITAE)

Axis	Control	ISE	IAE	ITAE
Yaw	LQR	0.0127	0.1491	1.4123
	MPC	0.0088	0.1271	1.3886
	LQR-STSMC	**4.245e-5**	**0.0016**	**0.0111**
Pitch	LQR	0.0090	0.1402	1.2546
	MPC	0.0019	0.0633	0.6502
	LQR-STSMC	**3.848e-5**	**0.0018**	**0.0104**
Roll	LQR	0.0096	0.1577	1.6730
	MPC	0.0023	0.0656	0.7993
	LQR-STSMC	**4.969e-5**	**0.0016**	**0.0075**

The comparative analysis of error metrics presented in Table 3 quantitatively validates the system's robustness under time-varying parameters. A holistic review of the data reveals a clear performance hierarchy: while the MPC controller generally outperforms the classical LQR providing notable reductions in error magnitudes across Pitch and Roll axes the proposed LQR-STSMC architecture consistently achieves the superior performance across all metrics and axes. In terms of ISE, which reflects the severity of transient oscillations, the proposed method yields values in the order of 10^{-5} Yaw, indicating a drastic suppression of peak errors compared to both benchmarks. Similarly, the IAE and ITAE metrics, which indicate the cumulative error and settling speed, confirm the adaptability of the hybrid controller. The LQR-STSMC reduces these values to negligible levels.

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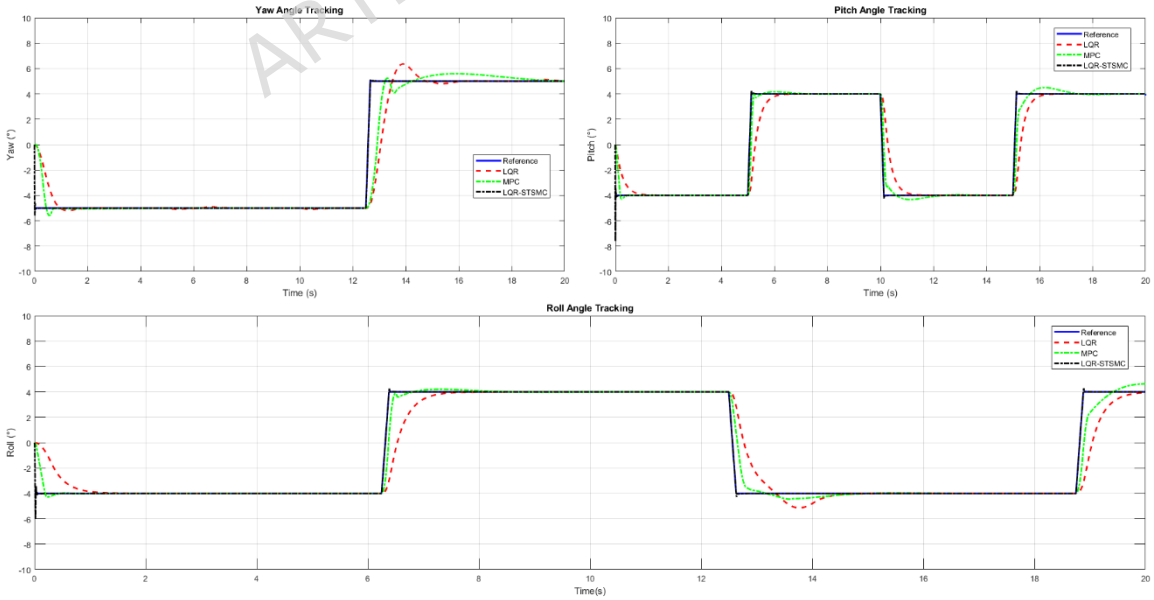

Figure 3. Time Responses of Attitude Angles Under Time-Varying Thrust Coefficients

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As shown in Figure 3, under time varying thrust conditions, the MPC controller exhibits the fastest rise time but fails to eliminate steady-state errors, resulting in worse long term tracking accuracy than even the classical LQR due to model mismatch. In contrast, the proposed LQR-STSMC maintains superior stability, effectively combining rapid transient response with robust error elimination to prevent the deviations observed in both benchmark methods.

Scenario 3: Motor Failure (Fault-Tolerant Control)

In this scenario, the fault-tolerant performance of the control system is evaluated under a simulated motor failure condition. A fault was modelled in which the second motor is completely disabled between 10 and 12 seconds of the simulation period. During this time, the corresponding motor produces no thrust, and its control input is set to zero.

Such sudden and temporary failures are commonly encountered in unmanned aerial vehicles, especially due to mechanical or electrical issues influenced by external environmental factors. This scenario was designed to analyze the ability of the proposed hybrid controller to maintain system stability and continue performing attitude tracking tasks in the presence of motor faults.

Table 4. Error Metrics (ISE, IAE, ITAE) of LQR, MPC and LQR-STSMC Controllers Under Motor Failure Condition

Axis	Control	ISE	IAE	ITAE
Yaw	LQR	0.0145	0.2082	2.0992
	MPC	0.0117	0.2381	2.6418
	LQR-STSMC	0.0025	0.0737	0.8103
Pitch	LQR	0.0497	0.4274	4.5205
	MPC	0.0638	0.3689	3.9687
	LQR-STSMC	**0.0527**	**0.3588**	**3.9376**
Roll	LQR	0.0095	0.1546	1.6298
	MPC	0.0023	0.0654	0.7967
	LQR-STSMC	**5.059e-5**	**0.0016**	**0.0075**

The performance metrics presented in Table 4 highlight the fault tolerance capabilities of the control strategies under a critical motor failure scenario. A general overview of the data reveals that the LQR-STSMC architecture maintains a decisive superiority, particularly in the Roll and Yaw axes, where it drastically suppresses tracking errors compared to both LQR and MPC benchmarks. Although the Pitch axis exhibits elevated error levels for all controllers due to the severe loss of thrust and physical actuation limits, the proposed hybrid method still demonstrates the highest resilience. By achieving the lowest cumulative error values across all axes, the LQR-STSMC confirms its ability to mitigate the destabilizing effects of actuator faults more effectively than conventional approaches.

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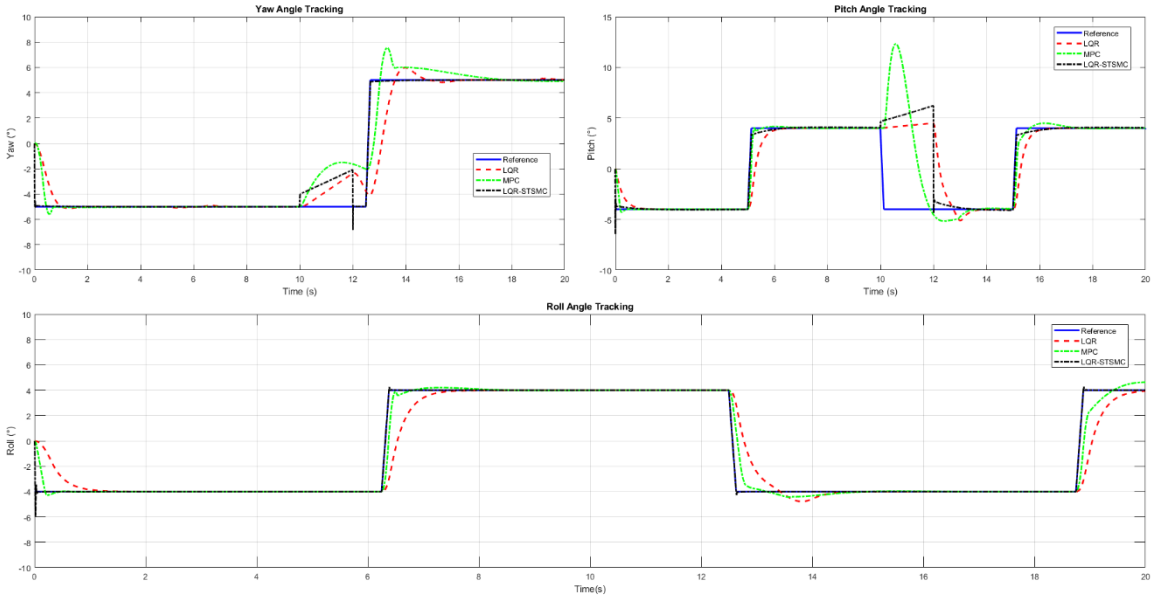

Figure 4. Time Responses of Attitude Angles (Yaw, Pitch, Roll) Under Motor Failure Scenario

Figure 4 illustrates the severe impact of motor failure on the system's stability. While the classical LQR controller suffers from significant deviations, particularly in the pitch axis, the MPC exhibits the highest overshoot amplitudes among all tested strategies, indicating an aggressive but unstable transient response to the actuator loss. In stark contrast, the proposed LQR-STSMC effectively mitigates these destabilizing effects, maintaining accurate tracking in yaw and roll axes. These results confirm the hybrid controller's superior adaptability and robustness against severe faults, making it highly suitable for safety-critical flight applications.

Scenario 4: Payload Variation (Load Drop Condition)

In this scenario, the effects of sudden mass variations such as those that may occur during payload transport missions on control performance are investigated. Initially, the system carries an additional 5 kg payload on top of its nominal mass. At 8 seconds, the payload is released, resulting in an abrupt decrease in the total system mass.

Such load changes are common in practice during the completion of delivery missions, emergency payload release procedures, or dynamic task requirements. Through the simulation conducted, the adaptation capability and stability performance of the proposed controllers were analyzed in response to this sudden mass variation.

Table 5. Error Metrics (ISE, IAE, ITAE) of LQR, MPC and LQR-STSMC Controllers Under Sudden Payload Variation

Axis	Control	ISE	IAE	ITAE
Yaw	LQR	0.0144	0.2010	1.5515
	MPC	**0.0131**	**0.2880**	**2.0523**
	LQR-STSMC	**8.811e-5**	**0.0062**	**0.0520**
Pitch	LQR	0.0101	0.1455	1.2749

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	MPC	**0.0070**	**0.1993**	**1.3012**
	LQR-STSMC	**0.0002**	**0.0390**	**0.3940**
Roll	LQR	0.0113	0.1730	1.7073
	MPC	**0.0074**	**0.1928**	**1.4748**
	LQR-STSMC	**8.308e-5**	**0.0026**	**0.0084**

The quantitative analysis of error metrics under sudden payload variation, as summarized in Table 5, reveals a critical insight into the controller's sensitivity to mass and inertia changes. While the MPC controller manages to reduce peak deviations (ISE) in certain axes compared to LQR, it notably struggles with accumulated errors (IAE and ITAE), occasionally performing worse than the linear baseline. This degradation in MPC performance is attributed to the discrepancy between the controller's internal model and the actual system dynamics following the payload change. In contrast, the LQR-STSMC architecture demonstrates remarkable insensitivity to these parameter variations. Across all axes, the proposed method yields error metrics that are orders of magnitude lower than both benchmarks, confirming that the sliding mode component effectively compensates for significant changes in the system's physical parameters without requiring model recalibration.

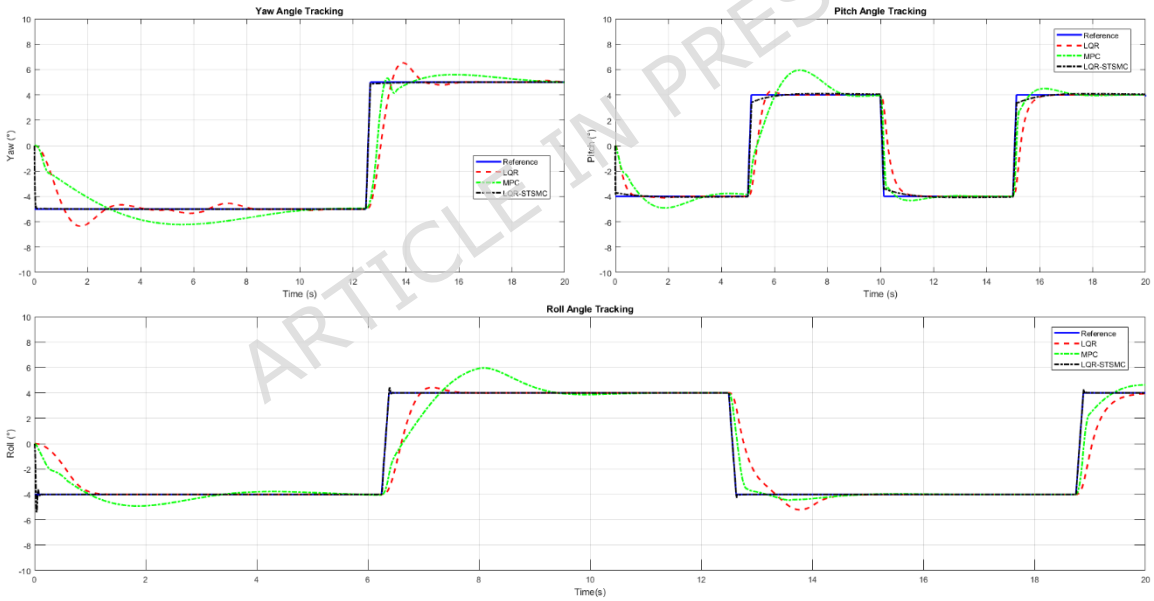

Figure 5. Time Responses of Attitude Angles (Yaw, Pitch, Roll) Under Sudden Payload Release Scenario

An examination of Figure 5 reveals distinct behavioural differences following the sudden payload release. Surprisingly, the MPC controller exhibits the poorest performance, displaying larger transient deviations and significant steady-state errors compared to even the classical LQR. This deterioration is attributed to the acute mismatch between the controller's internal model and the altered system dynamics (mass and inertia reduction). While the LQR controller also struggles with increased oscillations and a prolonged settling time, the proposed LQR-STSMC controller demonstrates superior robustness. It ensures rapid recovery

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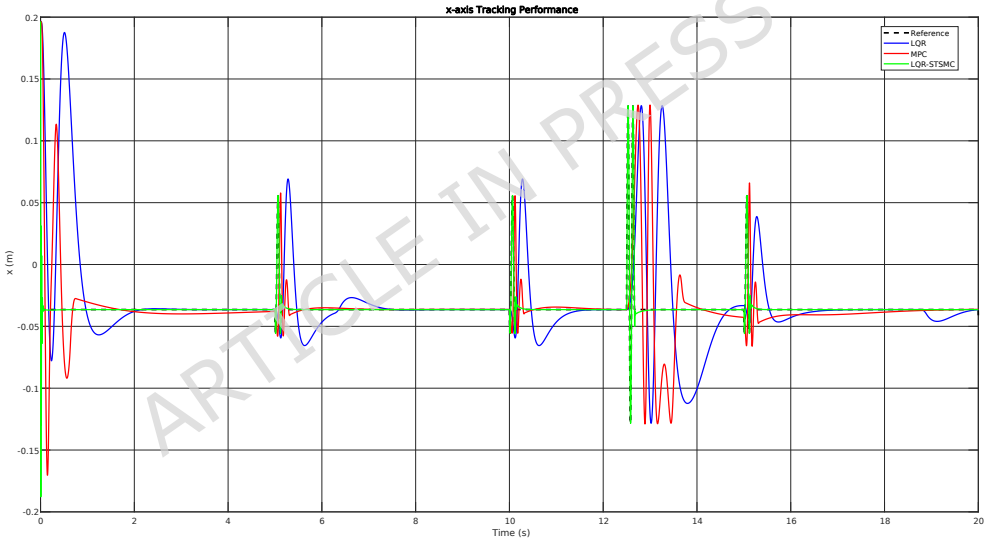
across yaw, pitch, and roll axes, allowing the system to converge to the reference values quickly with minimal oscillations, proving its insensitivity to sudden physical parameter changes.

3D Spatial Analysis of the Rotor Tip Trajectory

To visualize the stabilization performance in the Cartesian space, a kinematic transformation analysis was conducted. Since the Quanser 3-DOF Hover system is mechanically pivoted, a direct translational flight trajectory cannot be generated. However, the oscillation amplitude of the quadrotor arms provides a critical metric for stability. Using the kinematic relationship between the body frame and the inertial frame, the spatial trajectory of the X-axis rotor tip was reconstructed. Let L = 0.1965 m denote the distance from the pivot point to the rotor center. The position of the rotor tip in the body frame is defined as $P_{body} = [L, 0, 0]^T$. The corresponding coordinates in the inertial frame ($P_{inertial} = [x, y, z]^T$) are computed using the rotation matrix R: [57]

$$P_{inertial} = R_z(\varphi)R_y(\theta)R_x(\varnothing)P_{body} \quad (27)$$

Figure 6 illustrates the 3D spatial evolution of the rotor tip under LQR, MPC, and LQR-STSMC control strategies.



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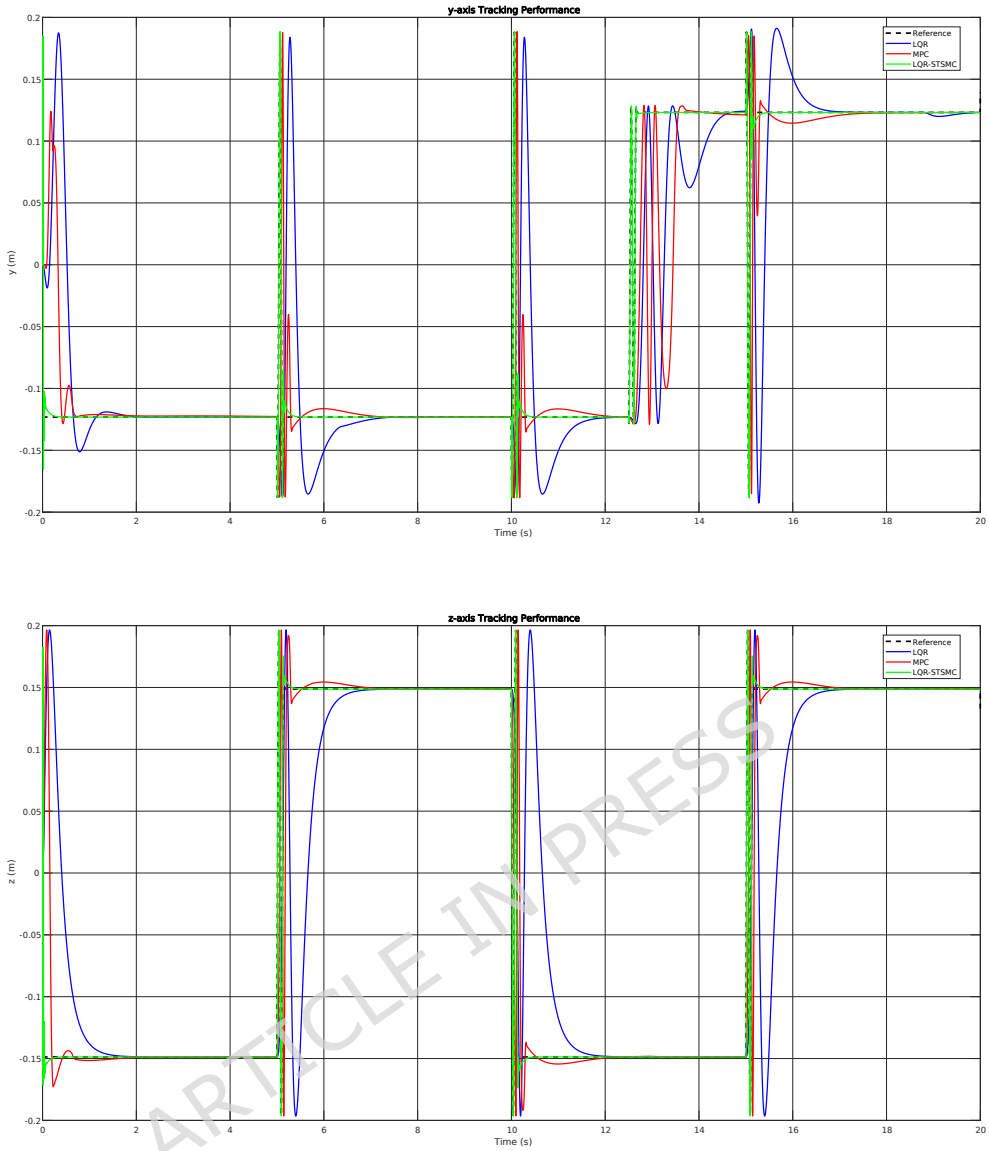

Figure 6. 3D Spatial Evolution of the Rotor Tip Under LQR, MPC, and LQR-STSMC Control Strategies

The results demonstrate that the proposed LQR-STSMC architecture significantly confines the rotor movement within a tighter spatial envelope compared to the benchmark controllers. This indicates that the hybrid controller effectively suppresses the mechanical vibrations and strictly maintains the end effector position near the equilibrium plane.

5. Discussion

Limitations of the Hybrid Framework While the proposed LQR-STSMC architecture significantly enhances tracking performance, it is subject to certain limitations regarding both theoretical design and physical constraints.

First, the LQR component relies on a linearized model, meaning its optimality is strictly valid near the equilibrium point. Second, the auxiliary

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STSMC signal depends on state derivatives, which can amplify measurement noise in real-world applications without proper filtering.

Crucially, the system is also bound by physical actuator limitations. In this study, a saturation block is implemented to restrict control signals to positive voltages with a maximum limit of 24 V, reflecting the actual hardware capacity of the motors. Furthermore, a rate limiter restricts setpoint step commands to 60 deg/s to ensure smoother control signals and prevent abrupt actuator saturation. Consequently, the controller's performance is bounded by these physical limits, particularly under conditions requiring control efforts that exceed the 24 V capacity or demand angular rates faster than 60 deg/s.

Finally, the mathematical model employed in this study assumes a rigid body structure and simplifies complex high order aerodynamic phenomena, such as blade flapping and ground effects. While these unmodeled dynamics have minimal impact during hover and stabilization tasks, they may introduce minor discrepancies between simulation and experimental results during highly aggressive maneuvers.

6. Conclusions

In this study, a hybrid control architecture, LQR-STSMC, was proposed and rigorously evaluated for the attitude stabilization of the Quanser 3-DOF Hover system. Additionally, the BB-BC optimization algorithm was employed to systematically tune the STSMC parameters, ensuring optimal disturbance rejection capability without manual tuning bias. To demonstrate the effectiveness of the proposed strategy, a comprehensive comparative analysis was conducted against the classical LQR and the optimization-based MPC under various scenarios, including wind disturbances, motor failures, and sudden payload variations.

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The quantitative results clearly demonstrate that the LQR-STSMC controller outperforms both benchmarks across all attitude axes (Yaw, Pitch, and Roll). While the MPC controller exhibited rapid initial responses, it suffered from significant steady-state errors and performance degradation under parametric uncertainties (such as payload release) due to model mismatch. In contrast, the proposed hybrid method maintained exceptional robustness. The performance metrics (ISE, IAE, and ITAE) indicate that the LQR-STSMC reduces tracking errors by orders of magnitude compared to both LQR and MPC, effectively suppressing both transient oscillations and long-term deviations.

Furthermore, the virtual 3D spatial analysis of the rotor tip trajectory confirmed the practical implications of these findings. The kinematic reconstruction showed that the superior attitude tracking provided by the LQR-STSMC directly translates into a tighter spatial envelope, significantly minimizing the physical oscillation and virtual positional drift of the system arms compared to the benchmarks.

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The proposed hybrid structure successfully combines the nominal stability of linear control with the chattering free robustness of the STSMC. It offers a reliable solution to common challenges such as actuator saturation and severe dynamic faults, where conventional linear and model-based methods fall short. Consequently, the LQR-STSMC framework presents a safe and high-precision alternative for flight control systems, robotic platforms, and payload carrying UAVs operating in uncertain environments.

For future work, the proposed hybrid controller will be primarily tested on the physical Quanser 3-DOF Hover platform to validate its real-time applicability. A key focus of this experimental phase will be addressing practical implementation challenges, particularly the mitigation of sensor noise which can be amplified by the derivative terms in the sliding surface and managing the computational load of the control algorithm on limited resource embedded processors. Following this, the method will be extended to a free flying 6-DOF quadrotor to verify its performance in fully unconstrained environments. Additionally, to address parameter tuning complexities, integration with optimization algorithms or Reinforcement Learning techniques is planned to auto tune the LQR and STSMC parameters. Finally, multi-agent flight missions and coordinated control scenarios are identified as potential long-term extensions of this research.

Competing interests
The author(s) declare no competing interests.

Funding Declaration
The study was not funded.

Data Availability Statement
The data has been generated during the study and will be available on reasonable request to the corresponding author.

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Contributions
S.B. conceived the study, wrote the main manuscript, developed the simulations, validated the results, and conducted the robustness analysis. C.S. and A.D. reviewed and revised the manuscript and contributed to the validation of the results. All authors have read and approved the final manuscript.

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